CHAPTER

10

LOW-RISK HUMAN PAPILLOMAVIRUS: GENITAL WARTS, CANCER AND RESPIRATORY PAPILLOMATOSIS

Charles J.N. Lacey[1], Nuria Guimera[2] and Suzanne M. Garland[3]

[1]York Biomedical Research Institute, University of York, York, United Kingdom [2]DDL Diagnostic Laboratory, Rijswijk, The Netherlands [3]Department of Obstetrics and Gynaecology, University of Melbourne, VIC, Australia

THE PROBLEM: GENITAL WARTS AND LARYNGEAL PAPILLOMATOSIS

Genital warts can also be referred to as anogenital warts and condylomata acuminata. In this chapter, we will use the term Genital Warts and the abbreviation GW for consistency. GW usually present as small papillomas/warty growths on the external anogenital skin. They also occur but less frequently on internal anogenital mucosae, that is, the cervix, vagina, anal canal, and urethral meatus. GW do not usually result in major morbidity or mortality. However, they occasionally persist for long periods and rarely such longstanding infection may progress to malignancy. Thus, ~2% of external genital cancers, namely anal, vulval, and penile cancers, are caused by HPV 6 or HPV 11 (discussed later). However, GW are usually perceived as unsightly and disfiguring by the infected person, and are associated with psychological morbidity and feelings of shame [1].

Therefore, many persons with GW will seek treatment for their lesions, and this can be in various healthcare settings depending on local services. However, treatment is by no means straightforward, as there are many therapies in use, and these are not always used in a logical and efficient way, and inadequate response and recurrence after apparent clearance are frequently seen. Indeed, some years ago we showed that in a setting of unstructured treatment practices, after 3 months of treatment and follow-up, 44% of men and 38% of women still had lesions [2]. This is not only a problem for the infected subjects but also a problem for the service providers, as large amounts of time and money are diverted to the management of GW, which in turn detracts from efforts to manage other sexually transmitted diseases (STDs). One estimate from Australia put the annual healthcare costs of GW at >A$14 million per year [3].

Laryngeal papillomatosis is a disease characterized by the growth of multiple papillomas, usually arising from the larynx. The terms laryngeal papillomatosis and recurrent respiratory papillomatosis (RRP) are both used with overlapping meanings. In this chapter, we will mainly use the term laryngeal papillomatosis for consistency and convenience. The majority of patients present with hoarseness or stridor due to the growth of papillomas on the vocal cords or in the larynx.

Human Papillomavirus. DOI: https://doi.org/10.1016/B978-0-12-814457-2.00010-6
© 2020 Elsevier Inc. All rights reserved.

Young age at onset (<3 years), and infection by HPV 11 is associated with increased severity of the disease, as measured by number of required surgical procedures, severity of hoarseness, or airway obstruction [4,5]. The natural history of the disease is highly variable and can either persist for many years, be interspersed with short or long periods of remission, or even result in mortality. Recurrence rates vary between patients, with the worst cases requiring surgery under general anesthesia as frequently as every 3 weeks to maintain an airway. Respiratory papillomas are primarily located in the larynx, but approximately 17% of patients have tracheal disease and 5% pulmonary involvement. The papillomas are usually benign, but malignant conversion occurs in ~2% of cases [6].

HISTORICAL ASPECTS OF GENITAL WARTS AND LARYNGEAL PAPILLOMATOSIS

GW were well known to the Greeks and the Romans, and a number of authors already considered them an STD [7]. The first clear proposal in the modern era that a transmissible agent was responsible for the development of warts was given in 1824 by Sir Astley Cooper [8]. He stated that warts "frequently secrete a matter which is able to produce a similar disease in others." He cites two examples of this, the first being the accidental injury of a surgical dresser with a scalpel that had just been used for the removal of warts, and with warts developing at the inoculation site within a short time. The second example was a case of sexual transmission of GW from husband to wife. In 1897 Cathcart provided an excellent account of GW in which he proposed that they were a specific disease independent of other genital pathology, that the lesions were benign neoplasms that were contagious, and he cites case histories of two pairs of sexual partners to support this [9].

Barrett et al. (1954) were the first to publish direct evidence supporting sexual transmission in the acquisition of GW [10]. They reported on 66 soldiers returning from the Korean War who all had sexual contact overseas and had developed GW, with 38% of the subjects having another concurrent STD. They also described 24 wives of these returning soldiers who presented with GW usually between 4 and 6 weeks after their husbands' return, and in whom 22 of the husbands had concurrent GW and 2 a recent history of such lesions. A number of authors reported attempt to demonstrate viral particles by electron microscopy of specimens of GW in the mid-1960s, but the most convincing findings were those of Oriel and Almeida (1970) in which 13 out of 25 genital wart specimens were shown to contain viral particles [11].

In 1971 Oriel's seminal work, "Natural history of genital warts," was published [12]. He reported on 332 men and women with GW. He showed that the age distribution of these subjects was virtually identical to that of gonorrhea, and that the one-third of the subjects had other concurrent sexually transmitted infections (STIs). In an assiduous exercise in contact tracing, Oriel showed that the incubation period ranged between 3 weeks and 8 months with an average of 12 weeks, and that the infectivity rate from an index case was approximately 65%. The detailed nature of his observations had a profound impact in both Europe and the United States, and brought about the unequivocal acceptance of GW as an STD.

Shortly after this in 1974, Harald zur Hausen and Lutz Gissmann began their work on human papillomavirus (HPV) as a cause of different anogenital lesions, based on anecdotal reports of the

rare malignant conversion of GW into squamous cell carcinomas. In 1980 low-risk HPVs were the first genotypes isolated: HPV 6 from GW and HPV 11 from laryngeal papillomas [13,14]. HPV 6 was identified both in typical GW and also in locally invasive giant condyloma acuminata. Later, using HPV 11 as a probe, it was possible to isolate HPV 16 from cervical cancer biopsies. In 2008 Harald zur Hausen was awarded the Nobel Prize in Physiology or Medicine for this discovery and his pioneering work in this area.

Laryngeal papillomatosis was superlatively described in 1871 by the pioneering British otolaryngologist Morell Mackenzie [15]. He acknowledges probably the earliest description by Koderick in 1750, a series of 31 cases by Ehrmann in 1850, and then the first direct in vivo visualization by Czernak, the inventor of indirect laryngoscopy using a specially designed mirror. Mackenzie makes a number of startling observations, reporting 67 cases of papillomatosis, describes their histology being "of warty character" and draws parallels with GW, and reports that the lesions contain what are now referred to as koilocytes, with "dense nuclei surrounded by a halo." Following this laryngeal papillomatosis was recognized as a rare disease that was often difficult to treat. It was not until the "golden era" of HPV research that significant further advances in the understanding of RRP were made. In 1982 Phoebe Mounts, Keerti Shah, and Haskins Kashima from John Hopkins showed both HPV antigens and the presence of the recently described HPV 6 in a series of laryngeal papillomas [16]. The following year Betty Steinberg, Allan Abramson, and colleagues from the Long Island Jewish Medical Centre showed the presence of latent laryngeal HPV in RRP cases during clinical remission [17].

RISK FACTORS, NATURAL HISTORY, AND TREATMENT OF GENITAL WARTS AND LARYNGEAL PAPILLOMATOSIS

GW are quite frequently associated with other STIs. We examined the demographic and geospatial risk factors for GW and their interrelation with other STIs in an urban area in the United Kingdom [18]. Regression analysis showed that young age (15–24 years), ethnicity (Black > White > Asian), and social deprivation were independent risk factors for all STDs. There were highly significant correlations in the geospatial distribution of STD incidence rates, and GW and chlamydia had the widest geospatial distribution.

The natural history of untreated GW is poorly understood, as such studies would be unethical. Some cases of GW do undergo spontaneous resolution, and this is known to be associated with the development of a cell-mediated immune response [19]. Even with modern treatments, the average duration of an episode of GW is ~3 months but with wide variation around the mean [20].

There are a variety of medical treatments for GW and various national guidelines for their use. Some of these contain algorithms to determine treatment and treatment sequence, and patients and physicians can choose between patient-applied, multiple session physician-delivered, and single-session physician-delivered using local anesthesia options [21].

The strongest risk factor for juvenile-onset laryngeal papillomatosis is a maternal history of GW in pregnancy which is associated with a >200 times increased risk for the infant, and a "classical case" is a firstborn child delivered vaginally to a teenage mother [22]. The risk of adult-onset laryngeal papillomatosis is associated with lifetime number of sexual partners and oro-genital sex [23].

Risk factors for developing RRP among children of women with GW are poorly understood. Patients have a widespread latent HPV infection in clinically normal tissues of the larynx, trachea, and bronchi. In latency the viral DNA is present but with essentially no expression of viral RNA and no clinical or histologic evidence of disease. Recurrent papillomas are thought to be due to repeated activation of the latent infection, rather than reinfection or "seeding" of the virus during surgery, as was originally thought by many physicians. However, a significant fraction of the normal population carries latent HPV DNA in their airway but with no history of the disease, suggesting additional factors in pathogenesis.

Many therapies have been tried for RRP, with limited success and often with significant side effects. These include various surgical approaches, topical and systemic treatment with immunomodulators, antivirals, and chemotherapeutic drugs. Unfortunately, with the exception of the two large interferon studies, most reports have been either case reports or small clinical series with no controls, and at present, there are still no proven effective adjuvant therapies in addition to surgical ablation.

PATHOLOGY OF GENITAL WARTS AND LARYNGEAL PAPILLOMATOSIS

GW are usually multiple and multifocal, occurring in the vulval, penile, perianal and vaginal regions, and less commonly the cervix. HPV infects long-lived cells (probably stem cells) on the basal cell layer of the epithelium, and then undergoes its life cycle in differentiation-committed cells as they migrate upwards within the epithelium [24]. Fig. 10.1 illustrates some of the differences between low-risk and high-risk HPV lesions. Typical histological features of a genital wart include elongation of the dermal papillae (papillomatosis), hyperplasia of the stratum spinosum (acanthosis), and in the stratum granulosum, large, vacuolated cells (koilocytes) are seen (Fig. 10.2). The histology of laryngeal papillomatosis lesions is essentially the same as that of GW lesions. There may often be some features of intraepithelial neoplasia within GW lesions. Such features include a failure of orderly maturation, excessive and abnormal mitotic activity, and cells with an increased nuclear-to-cytoplasmic ratio. There are still concurrent terminologies to describe such intraepithelial neoplasia changes (i.e., IN1, IN2, IN3) and the SIL classification [low-grade squamous intraepithelial lesion (LSIL), high-grade squamous intraepithelial lesion (HSIL)].

At all anogenital sites, GW may be misdiagnosed as an LSIL. For some authors, these terms are interchangeable. If GW present moderate cytological atypia and a dysplastic growth pattern, the lesions are classified as IN 2/3 or HSIL, but usually in those cases a high-risk HPV is also present. Other differential diagnoses of GW are squamous papilloma, giant condyloma, and verrucous carcinoma. Therefore defining the role of HPVs 6 and 11 in these lesions may assist in accurate diagnosis.

A rare variant of GW is the borderline malignant tumor referred to as giant condyloma or Buschke–Lowenstein tumor. These can arise at any anogenital site and can be associated with various immunodeficiency states. Giant condyloma is induced by HPVs 6 and 11 and cause local destruction without metastatic spread. Chemoradiotherapy may be the best treatment option [25].

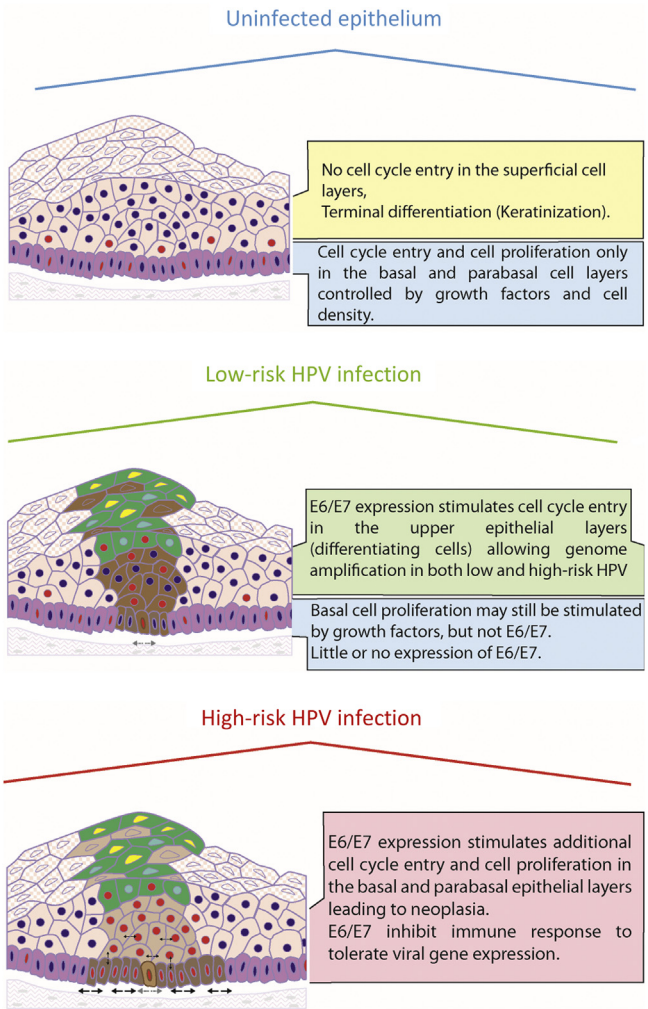


FIGURE 10.1

Explaining the different cellular differentiation states related to viral protein expression in lesions caused by low-risk and in high-risk HPV.

Adapted from Egawa N, Doorbar J. The low-risk papillomaviruses. Virus Res 2017;231:119–27.

HUMAN PAPILLOMAVIRUS IN GENITAL WARTS AND LARYNGEAL PAPILLOMATOSIS

GW and laryngeal papillomatosis are exophytic lesions arising from productive viral infection by HPV 6 or HPV 11, and in the most highly controlled studies either HPV 6 or HPV 11 are found within the lesions at rates close to 100% [26,27]. HPVs 6 and 11 are exceedingly common in

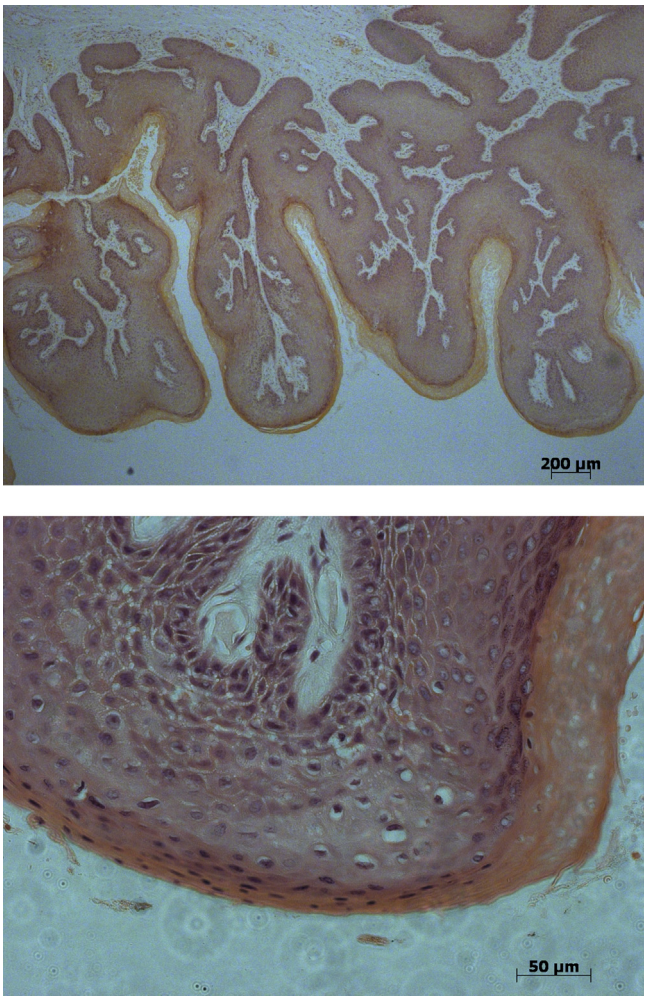
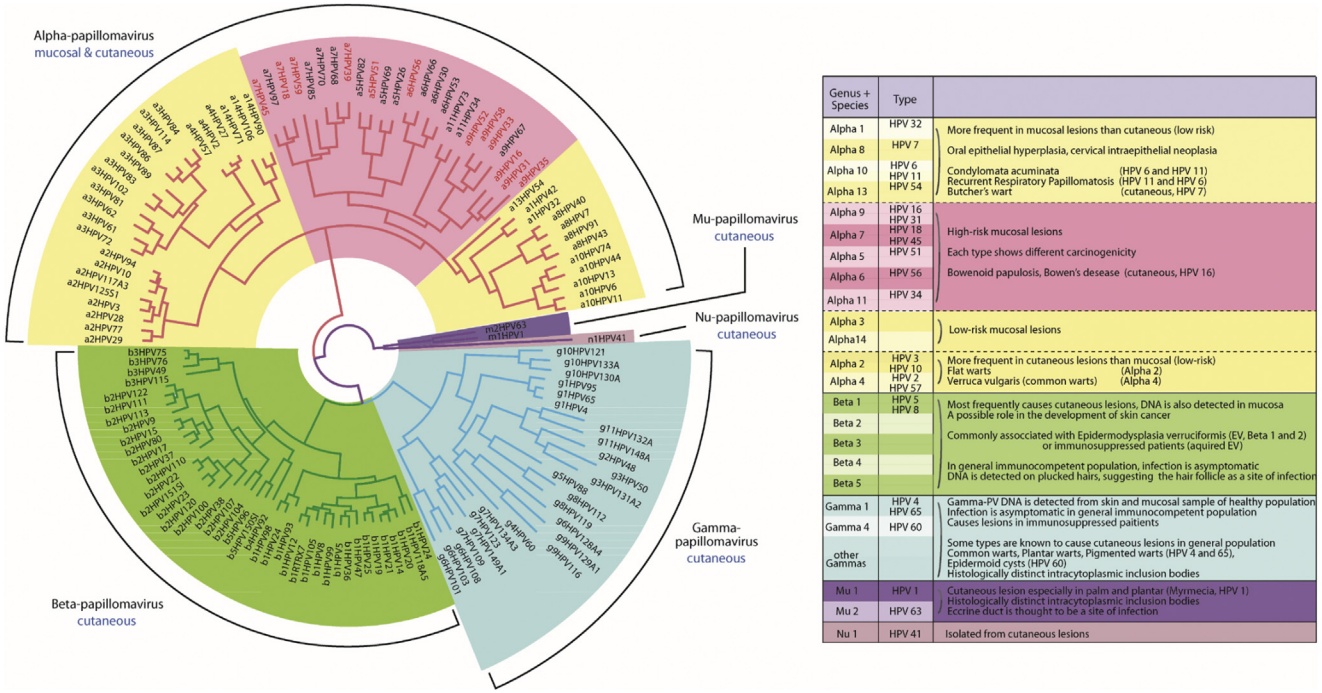


FIGURE 10.2

H&E sections of a typical GW (Lacey original).

benign HPV mucosal lesions and rarely detected in HPV-associated cancers, and therefore considered low-risk HPVs, where risk implies the potential for malignant transformation [24]. They are part of a distinct taxonomic HPV species, Alpha 10 (Fig. 10.3). The prevalence of different HPV types differs by type of lesion. HPV 6 is more prevalent than HPV 11 in GW, whereas HPVs 6 and 11 are found at similar frequencies in laryngeal papillomatosis [26–28].

HPV is usually present as a single infection and less frequently as multiple infections, which arise as separate lesional clones. The prevalence of multiple HPV infections decreases with increasing severity of the lesion, being 50%–80% in anogenital warts and 12% in cervical cancer.



Genus + Species	Type	
Alpha 1	HPV 32	More frequent in mucosal lesions than cutaneous (low risk)
Alpha 8	HPV 7	Oral epithelial hyperplasia, cervical intraepithelial neoplasia
Alpha 10	HPV 6	Condylomata acuminata (HPV 6 and HPV 11)
Alpha 10	HPV 11	Recurrent Respiratory Papillomatosis (HPV 11 and HPV 6)
Alpha 13	HPV 54	Butcher's wart (cutaneous, HPV 7)
Alpha 9	HPV 16	High-risk mucosal lesions Each type shows different carcinogenicity Bowenoid papulosis, Bowen's desease (cutaneous, HPV 16)
Alpha 9	HPV 31	
Alpha 7	HPV 18	
Alpha 7	HPV 45	
Alpha 5	HPV 51	
Alpha 6	HPV 56	
Alpha 11	HPV 34	
Alpha 3		Low-risk mucosal lesions
Alpha14		
Alpha 2	HPV 3	More frequent in cutaneous lesions than mucosal (low-risk)
Alpha 2	HPV 10	Flat warts (Alpha 2)
Alpha 4	HPV 2	Verruca vulgaris (common warts) (Alpha 4)
Alpha 4	HPV 57	
Beta 1	HPV 5	Most frequently causes cutaneous lesions, DNA is also detected in mucosa A possible role in the development of skin cancer Commonly assocciated with Epidermodysplasia verruciformis (EV, Beta 1 and 2) or immunosuppressed patients (aquired EV) In general immunocompetent population, infection is asymptomatic DNA is detected on plucked hairs, suggesting the hair follicle as a site of infection
Beta 1	HPV 8	
Beta 2		
Beta 3		
Beta 4		
Beta 5		
Gamma 1	HPV 4	Gamma-PV DNA is detected from skin and mucosal sample of healthy population Infection is asymptomatic in general immunocompetent population Causes lesions in immunosuppressed paitients Some types are known to cause cutaneous lesions in general population Common warts, Plantar warts, Pigmented warts (HPV 4 and 65), Epidermoid cysts (HPV 60) Histologically distinct intracytoplasmic inclusion bodies
Gamma 1	HPV 65	
Gamma 4	HPV 60	
other Gammas		
Mu 1	HPV 1	Cutaneous lesion especially in palm and plantar (Myrmecia, HPV 1) Histologically distinct intracytoplasmic inclusion bodies Eccrine duct is thought to be a site of infection
Mu 2	HPV 63	
Nu 1	HPV 41	Isolated from cutaneous lesions

FIGURE 10.3

Showing the taxonomy and evolutionary relationships between HPVs.

Reproduced from Egawa N, Doorbar J. The low-risk papillomaviruses. Virus Res 2017;231:119–27.

Multiple infections with other HPVs include both low-risk and high-risk HPV types [26]. The prevalence of multiple infections varies with the HPV detection method used in the different studies, as not all tests have same specificity and sensitivity in the detection of the different HPV types.

Multiple infections and simultaneous lesions are common in anogenital sites prone to infection. Studies have shown how cervical HPV infections act independently in the development of preneoplastic and neoplastic lesions in the cervix and that each lesion present in a patient is the result of a specific HPV type [29,30]. Simultaneous anogenital lesions can be driven by one or more HPV types, may originate at the same or different times, and may progress independently. The high prevalence of multiple infections detected in some GW creates difficulties in the study of their natural history. In a recent study of penile lesions in men presenting with penile cancer, we found atypical GW with high-risk HPV 16 and p16 biomarker expression and in the same patient invasive carcinoma and high-grade precancers with the same high-risk HPV 16 and p16 expression pattern. This finding suggests a possible, rare precursor role of the infrequent HPV 16/p16 positive GW [31]. In laryngeal papillomatosis the presence of multiple infections is infrequent and simultaneous precancerous lesions rare [5].

LOW-RISK HUMAN PAPILLOMAVIRUS TYPES AND CANCER

HPVs are phylogenetically classified in different genera (e.g., α, β, γ, μ, ν—Fig. 10.3). There is a strong concordance between HPV phylogeny, viral natural history, and carcinogenicity. In 2003 an epidemiologic risk classification for HPV was published [32], classifying HPVs as high-risk or low-risk based on the strength of the association in a metaanalysis of each type with cervical cancer. Subsequently, the epidemiological risk classification was compared with phylogenetic relationships of DNA and amino acid sequences. Evolutionary models clustered all α-HPVs into three ancestral branches [33]. The first and third ancestral branches include low-risk HPVs (α1, α2, α3, α4, α8, α10, α13, and α14 species) and the second ancestral branch high-risk HPVs (α5, α6, α7, α9, α12, and α11 species). These findings suggested that the carcinogenic potential of α-genera HPV genotypes is closely related to their species classification.

However, contrary to their biological description as “low-risk” or “noncarcinogenic” viral genotypes, low-risk HPVs have been identified in anogenital and head and neck cancers, although with a low frequency. The prevalence of low-risk HPVs decreases with increasing severity of the lesion, reflecting the lower oncogenic potential and more frequent regression of the lesions. Low-risk HPVs 6, 11, 40, 42, and 44 have been detected in 1.7% of cervical carcinomas as single and/or multiple genotypes (0.27% and 1.4%, respectively) [34]. Low-risk HPVs 6 and 11 have a prevalence of 2% and 1.3%, respectively, in HSIL and 6.2% and 2.9%, respectively, in LSIL [35].

At the Catalan Institute of Oncology, we performed several large epidemiological studies which included 13,328 HPV-related malignant carcinomas from different anogenital sites, such as cervical, vaginal, vulvar, anal, and penile. These studies were the basis to investigate the presence of low-risk HPVs in malignant carcinomas as a large number of patients are needed to detect these rare cases. Only 57 cases showed single low-risk HPV infection [30]. Low-risk

HPV DNA was confirmed as a single infection in tumor cells in 56% (32/57) by laser capture microdissection followed by human papillomavirus polymerase chain reaction (HPV PCR) detection. All anogenital tumors with confirmed low-risk HPVs 6 and 11 showed verrucopapillary, well-differentiated, squamous, or transitional histology without p16 expression. HPVs 42 and 70 were associated with typical squamous carcinomas with p16 expression. These data support the causal involvement of low-risk HPVs in the carcinogenesis of $<2\%$ of anogenital malignancies [30].

Malignant transformation can occur in RRP but is infrequent, of the order of 2% [6,36]. Such cancer development is strongly linked with a long history, HPV 11, and involvement of the lung parenchyma. The HPV 11 genome can be integrated into such cases, suggesting deregulated viral gene expression followed by the acquisition of additional genetic and epigenetic modifications as seen for the α HR types [24].

THERAPEUTIC AND PROPHYLACTIC VACCINES AGAINST GENITAL WARTS AND LARYNGEAL PAPILLOMATOSIS

Therapeutic vaccination as a treatment for GW using poorly characterized autogenous wart preparations was first described in 1925, and similar therapy was also used in RRP [37,38]. Such approaches to treatment using autogenous vaccines were occasionally used for many years, but a double-blind controlled cross-over study versus placebo showed no differences in outcome [39]. We developed an HPV 6 L2E7 fusion protein therapeutic vaccine for GW, based on the findings of Jarrett & Campo in the BPV model, but a series of three trials showed no additional benefit over conventional therapy [40].

The foundation stone for HPV VLP vaccine development was the description by Jian Zhou and colleagues of the expression of HPV 16 L1 and its spontaneous assembly into VLPs in 1991 [41]. The subsequent development of the HPV VLP vaccines is fully covered in Chapters 15–17, but some of the key observations at this time were in HPV 6/11 systems. Rose and colleagues inoculated rabbits with HPV 11 VLPs, and using the Kreider athymic mouse model showed that vaccination-induced neutralizing antibodies against HPV 11 blocked infection [42]. Roden et al. assessed the serological relatedness of genital HPVs and showed that even in highly related HPV genotypes, such as HPVs 6 and 11, where amino acid sequence homology in L1 was $>85\%$, there was relatively limited cross-reactivity by neutralizing antibodies, which suggested that protection induced by HPV VLPs would be predominantly type specific [43]. Lowe et al. immunized African green monkeys with HPV 11 VLPs and showed that high levels of neutralizing antibodies were induced in serum, but also that significant levels of neutralizing antibodies could be detected via transudation in cervico-vaginal secretions [44]. The development of VLP vaccines then proceeded apace and culminated in 2006 with the licensing of Gardasil by the FDA. This was followed in 2007 with a number of key publications including that of Garland et al. demonstrating 100% efficacy of the HPV 6/11/16/18 vaccine in the per protocol population against all external anogenital and vaginal lesions, and all HPV 6/11 lesions [45]. These data triggered the widespread introduction of HPV VLP vaccination across the world.

PAST, PRESENT, AND FUTURE EPIDEMIOLOGY OF GENITAL WARTS AND LARYNGEAL PAPILLOMATOSIS

GW were a significant public health problem worldwide with rates of 0.1%–0.2% in general populations for both incidence of new cases, as well as prevalence/recurrent cases, before the introduction of HPV vaccination [46]. The UK National Survey of Sexual Attitudes and Lifestyles conducted in the year 2000 was a stratified population-based survey of 11,161 men and women aged 16–44 years. This found that 3.6% of men and 4.1% of women reported ever being diagnosed with GW, which were the commonest reported STI [47].

Incidence data for laryngeal papillomatosis in the prevaccine era are sparse but show greater variation between countries than that for GW. Incidence rates of 3.6, 4.3, and 0.24 (Denmark, the United States, Canada) per 10^6 person-years (py) in children, and 3.9 and 1.8 (Denmark, the United States) per 10^6 py for adult disease have been reported [48–50]. Juvenile-onset disease usually presents between the ages of 1 and 4 years and is equally distributed between males and females. Adult-onset disease has a broad peak between ages 20 and 40 years, with a male:female ratio of ~2:1 [22].

Rapid licensing of the HPV vaccines occurred in many countries from 2007 onwards. Economic modeling of the introduction of HPV vaccines confirmed their cost-effectiveness [51]. Australia was one the first countries to introduce a national state-funded HPV vaccination program in 2007. In Australia during 2007–09, an estimated 83% of females aged 12–17 years received at least one dose of the HPV vaccine and 70% completed the three-dose HPV vaccination course [52], and this degree of coverage has been maintained since then. GW represent a short incubation period HPV disease, and so can act as a sentinel readout to assess the effectiveness of HPV vaccination at a population level. Thus Donovan et al. observed that in sexual health clinics in Australia the rate of diagnosis of GW in young female residents decreased by 59% in 2007–09, that there was also a significant decrease (28%) in GW in nonvaccinated young heterosexual men, whereas no decline was seen in female nonresidents, older women >26 years, and men who have sex with men (MSM) [53].

These early signals of HPV 6/11 disease reduction were conclusively confirmed by the meta-analysis of Drolet et al. [54]. They identified 20 eligible studies conducted in 9 high-income countries representing 140 million py of follow-up. In countries with >50% female vaccination coverage, GW declined by 61% (RR 0.39, 0.22–0.71) in girls aged 13–19 years through vaccination, and in boys <20 years by 34% (0.66, 0.47–0.91) due to herd immunity. Australia extended qHPV vaccination to boys in 2013. Recent Australian data confirm the profound on-going decline in GW, with a 74% reduction in *all* females and a 65% reduction in *all* males [55]. Other countries with lesser degrees of HPV vaccine population coverage are also reporting ongoing declines in GW (United States – [56], Sweden – [57]). Recent modeling has investigated the likely course of GW epidemiology in Australia [58]. These data predict a 95% reduction in GW cases by 2060 in all the subgroups of females, males, and MSM, and by then ~50% of cases will be occurring in non-Australian born nonvaccinated travellers.

Surveillance for juvenile-onset RRP was established in Australia in 2012, and the data until end-2016 have been reported [59]. This found that the average annual incidence rates have declined significantly from 0.16 in 2012 to 0.02 in 2016. Of the incident cases, no mothers were vaccinated prepregnancy, 20% had a maternal history of GW, and 60% were firstborn. This was the first report

of a decline in laryngeal papillomatosis incidence in children following the introduction of the HPV vaccine program. A similar national-level surveillance program for pediatric RRP was established in Canada in 2007 at the time of HPV vaccine introduction, and recent data indicate that there have been a ~65% decrease in incidence since then compared to the period 1992–2016 [60].

CONCLUSION

GW and laryngeal papillomatosis are ancient historical diseases almost exclusively caused by HPVs 6 and 11. These diseases were associated with very significant physical and psychological morbidity and healthcare costs. In many countries a distinct increase in incidence and prevalence of GW was recorded in the period 1970–2000 thought to be associated with changes in sexual behavior. The astonishing HPV vaccine efficacy against HPV 6/11 observed in Phase 3 trials has led to the widespread uptake of the quadrivalent vaccine containing HPV 6/11 VLPs in many developed world countries. Such implementation from 2007 onwards can now be seen to be producing very significant reductions in GW and laryngeal papillomatosis disease incidence. This evidence strongly suggests that as HPV vaccine introduction increases worldwide, over the next 50 years, these diseases will become increasingly rare.

ACKNOWLEDGEMENTS

Thanks to Prof. Philip Quirke, University of Leeds, for histopathological advice, and Prof. John Doorbar, University of Cambridge, and Dr. Helen Ashwin, University of York, for assistance with the illustrations.

REFERENCES

[1] Maw RD, Reitano M, Roy M. An international survey of patients with genital warts: perceptions regarding treatment and impact on lifestyle. Int J STD AIDS 1998;9:571–8.
[2] Reynolds M, Murphy M, Waugh MA, Lacey CJN. An audit of treatment of genital warts: opening the feedback loop. Int J STD AIDS 1993;4:226–31.
[3] Pirotta M, Stein AN, Conway EL, Harrison C, Britt H, Garland S. Genital warts incidence and healthcare resource utilisation in Australia. Sex Transm Infect 2010;86:181–6.
[4] Wiatrak BJ, Wiatrak DW, Broker TR, Lewis L. Recurrent respiratory papillomatosis: a longitudinal study comparing severity associated with HPV types 6 and 11 and other risk factors in a large pediatric population. Laryngoscope 2004;114:1–23.
[5] Seedat RY, Thukane M, Jansen AC, Rossouw I, Goedhals D, Burt FJ. HPV types causing juvenile recurrent laryngeal papillomatosis in South Africa. Int J Pediatr Otorhinolaryngol 2010;74:255–9.
[6] Dedo HH, Yu KC. CO2 laser treatment in 244 patients with respiratory papillomas. Laryngoscope 2001;111:1639–44.
[7] Bafverstedt B. Condylomata acuminata - past and present. Acta Dermato-Venereol 1967;47:376–81.
[8] Cooper A. Surgical lectures, no 62: warts. Lancet 1824;3:336–7.
[9] Cathcart CW. Venereal warts, a contagious form of tumour. J Pathol Bacteriol 1897;4:160–72.

[10] Barrett TJ, Silbar JD, McGinley JP. Genital warts - a venereal disease. JAMA 1954;154:333–4.
[11] Oriel JD, Almeida JD. Demonstration of virus particles in human genital warts. Br J Vener Dis 1970;46:37–42.
[12] Oriel JD. Natural history of genital warts. Br J Vener Dis 1971;47:1–13.
[13] Gissmann L, de Villiers EM, zur Hausen H. Analysis of human genital warts (condylomata acuminata) and other genital tumors for human papillomavirus type 6 DNA. Int J Cancer 1982;29:143–6.
[14] Gissmann L, Diehl V, Schultz-Coulon HJ, zur Hausen H. Molecular cloning and characterization of human papilloma virus DNA derived from a laryngeal papilloma. J Virol 1982;44:393–400.
[15] Mackenzie M. Essays on growths in the larynx: with reports, and an analysis of one hundred consecutive cases. London: J &A Churchill; 1871 (Available via Google Books).
[16] Mounts P, Shah KV, Kashima H. Viral etiology of juvenile- and adult-onset squamous papilloma of the larynx. Proc Natl Acad Sci 1982;79:5425–9.
[17] Steinberg BM, Topp WC, Schneider PS, Abramson AL. Laryngeal papillomavirus infection during clinical remission. N Engl J Med 1983;308:1261–4.
[18] Monteiro EF, Lacey CJN, Merrick D. The interrelation of demographic and geospatial risk factors between four common sexually transmitted diseases. Sex Transm Inf 2005;81:41–6.
[19] Coleman N, Birley HDL, Renton AM, Hanna NF, Ryait BK, Byrne M, et al. Immunological events in regressing genital warts. Am J Clin Pathol 1994;102:768–74.
[20] Woodhall SC, Jit M, Cai C, Ramsey T, Zia S, Crouch S, et al. Cost of treatment and QALYs lost due to genital warts: data for the economic evaluation of HPV vaccines in the United Kingdom. Sex Transm Dis 2009;36:515–21.
[21] Lacey CJ, Woodhall SC, Wikstrom A, Ross J. European guideline for the management of anogenital warts. J Eur Acad Dermatol Venereol 2013;27:e263–70.
[22] Gillison ML, Alemany L, Snijders PJ, Chaturvedi A, Steinberg BM, Schwartz S, et al. Human papillomavirus and diseases of the upper airway: head and neck cancer and respiratory papillomatosis. Vaccine 2012;30(Suppl. 5):F34–54.
[23] Kashima HK, Shah F, Lyles A, Glackin R, Muhammad N, Turner L, et al. A comparison of risk factors in juvenile-onset and adult-onset recurrent respiratory papillomatosis. Laryngoscope 1992;102:9–13.
[24] Egawa N, Doorbar J. The low-risk papillomaviruses. Virus Research 2017;231:119–27.
[25] Armstrong N, Foley G, Wilson J, Finan P, Sebag-Montefiore D. Successful treatment of a large Buschke-Lowenstein tumour with chemo-radiotherapy. Int J STD AIDS 2009;20:732–4.
[26] Brown DR, Schroeder JM, Bryan JM, Stoler MH, Fife KH. Detection of multiple human papillomavirus types in condylomata acuminata lesions from otherwise healthy and immunosuppressed patients. J Clin Microbiol 1999;37:3316–22.
[27] Hawkins MG, Winder DM, Ball SL, Vaughan K, Sonnex C, Stanley MA, et al. Detection of specific HPV subtypes responsible for the pathogenesis of condylomata acuminata. Virol J 2013;10:137.
[28] Wiatrak BJ, Wiatrak DW, Broker TR, Lewis L. Recurrent respiratory papillomatosis: a longitudinal study comparing severity associated with human papilloma viral types 6 and 11 and other risk factors in a large pediatric population. Laryngoscope 2004;114(11Pt 2 Suppl. 104):1–23.
[29] Quint W, Jenkins D, Molijn A, Struijk L, van de Sandt M, Doorbar J, et al. One virus, one lesion--individual components of CIN lesions contain a specific HPV type. J Pathol 2012;227:62–71.
[30] Guimerà N, Lloveras B, Lindeman J, Alemany L, van de Sandt M, Alejo M, et al. The occasional role of low-risk human papillomaviruses 6, 11, 42, 44, and 70 in ano-genital carcinoma defined by laser capture microdissection/PCR methodology: results from a global study. Am J Surg Pathol 2013;37:1299–310.
[31] Fernández-Nestosa MJ, Guimerà N, Sanchez DF, Cañete-Portillo S, Velazquez EF, Jenkins D, et al. Human Papillomavirus (HPV) Genotypes in condylomas, intraepithelial neoplasia, and invasive carcinoma of the penis using laser capture microdissection (LCM)-PCR: a study of 191 lesions in 43 patients. Am J Surg Pathol 2017;41:820–32.

[32] Muñoz N, Bosch FX, de Sanjosé S, Herrero R, Castellsagué X, Shah KV, et al. Epidemiologic classification of human papillomavirus types associated with cervical cancer. N Engl J Med 2003;348:518–27.
[33] Schiffman M, Herrero R, Desalle R, Hildesheim A, Wacholder S, Rodriguez AC, et al. The carcinogenicity of human papillomavirus types reflects viral evolution. Virology 2005;337:76–84.
[34] De Sanjose S, Quint WG, Alemany L, Geraets DT, Klaustermeier JE, Lloveras B, et al. Human papillomavirus genotype attribution in invasive cervical cancer: a retrospective cross-sectional worldwide study. Lancet Oncol 2010;11:1048–56.
[35] HPV Information Centre, Statistics. <https://www.hpvcentre.net/datastatistics.php> [accessed 21.02.2019].
[36] Gerein V, Rastorguev E, Gerein J, Draf W, Schirren J. Incidence, age at onset, and potential reasons of malignant transformation in recurrent respiratory papillomatosis patients: 20 years experience. Otolaryngol Head Neck Surg 2005;132:392–4.
[37] Biberstein H. Versuche uber Immunotherapie der Warzen und Kondylome. Klin Wochenschr 1925;4:638–41.
[38] Stephens CB, Arnold GE, Butchko GM, Hardy CL. Autogenous vaccine treatment of juvenile laryngeal papillomatosis. Laryngoscope 1979;89:1689–96.
[39] Malison MD, Morris R, Jones LW. Autogenous vaccine therapy for condyloma acuminatum: a double-blind controlled study. Br J Vener Dis 1982;58:62–5.
[40] Vandepapeliere P, Barrasso R, Meijer CJLM, Walboomers JMM, Wettendorff M, Stanberry LR, et al. Randomised controlled trial of an adjuvanted HPV6 L2E7 vaccine: multiple HPV type infection of external anogenital warts and failure of therapeutic vaccination. J Infect Dis 2005;192:2099–107.
[41] Zhou J, Sun XY, Stenzel DJ, Frazer IH. Expression of vaccinia recombinant HPV 16 L1 and L2 ORF proteins in epithelial cells is sufficient for assembly of HPV virion-like particles. Virology 1991;1991 (185):251–7.
[42] Rose RC, Reichman RC, Bonnez W. Human papillomavirus (HPV) type 11 recombinant virus-like particles induce the formation of neutralizing antibodies and detect HPV-specific antibodies in human sera. J Gen Virol 1994;75:2075–9.
[43] Roden RBS, Hubbert NL, Kirnbauer R, Christensen ND, Lowy DR, Schiller JT. Assessment of the serological relatedness of genital human papillomaviruses by haemagglutination inhibition. J Virol 1996;70:3298–301.
[44] Lowe RS, Brown DR, Bryan JT, Cook JC, George HA, Hofmann KJ, et al. Human papillomavirus type 11 (HPV-11) neutralizing antibodies in the serum and genital mucosal secretions of African green monkeys immunized with HPV-11 virus-like particles expressed in yeast. J Infec Dis 1997;176:1141–5.
[45] Garland SM, Hernandez-Avila M, Wheeler CM, Perez G, Harper DM, Leodolter S, et al. Quadrivalent vaccine against human papillomavirus to prevent anogenital diseases. N Engl J Med 2007;356:1928–43.
[46] Forman D, de Martel C, Lacey CJ, Soerjomataram I, Lortet-Tieulent J, Bruni L, et al. Global burden of human papillomavirus and related diseases. Vaccine 2012;30(Suppl. 5):F12–21.
[47] Fenton KA, Korovessis C, Johnson AM, McCadden A, McManus S, Wellings K, et al. Sexual behaviour in Britain: reported sexually transmitted infections and prevalent genital Chlamydia trachomatis infection. Lancet 2001;358:1851–4.
[48] Lindeberg H, Elbrønd O. Laryngeal papillomas: the epidemiology in a Danish subpopulation 1965-1984. Clin Otolaryngol Allied Sci 1990;15:125–31.
[49] Derkay CS. Task force on recurrent respiratory papillomas. A preliminary report. Arch Otolaryngol Head Neck Surg 1995;121:1386–91.
[50] Campisi P, Hawkes M, Simpson K, Canadian Juvenile Onset Recurrent Respiratory Papillomatosis Working Group. The epidemiology of juvenile onset recurrent respiratory papillomatosis derived from a population level national database. Laryngoscope 2010;120:1233–45.
[51] Jit M, Choi YH, Edmunds JD. Economic evaluation of human papillomavirus vaccination in the UK. BMJ 2008;337:a769.

[52] Tabrizi SN, Brotherton JM, Kaldor JM, Skinner SR, Cummins E, Liu B, et al. Fall in human papillomavirus prevalence following a national vaccination programme. J Infec Dis 2012;206:1645–51.
[53] Donovan B, Franklin N, Guy R, Grulich AE, Regan DG, Ali H, et al. Quadrivalent human papillomavirus vaccination and trends in genital warts in Australia: analysis of national sentinel surveillance data. Lancet Infec Dis 2011;11:39–44.
[54] Drolet M, Benard E, Boily M-C, Ali H, Baandrup L, Bauer H, et al. Population-level impact and herd effects following human papillomavirus vaccination programmes: a systematic review and meta-analysis. Lancet Infec Dis 2015;15:565–80.
[55] Chow EP, Callander D, Fairley CK, Guy RJ, Regan DG, Grulich AE, et al. The continuing decline in genital warts 10 years into the national human papillomavirus vaccination programme: national sentinel surveillance data 2004-2016. IPVC 2018 Sydney.
[56] Flagg EW, Torrone EA. Declines in anogenital warts among age groups most likely to be impacted by human papillomavirus vaccination, United States 2006-2014. Am J Public Health 2018;108:112–19.
[57] Herweijer E, Ploner A, Sparen P. Substantially reduced incidence of genital warts in women and men six years after vaccine availability in Sweden. Vaccine 2018;36:1917–20.
[58] Khawar L, McGregor S, Machalek D, Regan D, Kaldor J, Donovan B, et al. When will we eliminate genital warts in Australia? IPVC 2018 Sydney; 2018.
[59] Novakovic D, Cheng ATL, Zurynski Y, Booy R, Walker PJ, Berkowitz R, et al. A prospective study of the Incidence of juvenile-onset recurrent respiratory papillomatosis after Implementation of a national HPV vaccination program. J Infect Dis 2018;217:208–12.
[60] Campisi P. Should we be screening for oral HPV-related diseases? IPVC 2018 Sydney.